

BBO capstone project presentation template

Develop a concise presentation that captures:

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

Objective

The assignment provides eight unknown functions. Each accepts between two and eight numerical inputs in the range 0 to 1 and returns a single output value. The internal form of each function is not visible; the only available information is the output returned for each submitted input. The objective is to identify, for each function, the input combination that produces the highest output, within a strict budget of one query per function per week across the programme.

Approach

The process is an iterative search loop, run identically every week. For each function: maintain a complete history of all inputs tried and outputs received; fit a predictive model to that history; evaluate a large set of candidate inputs against a selection rule; submit the strongest candidate; and incorporate the returned result into the history for the following round.

Around this core sits a governance layer developed over the twelve rounds: a weekly reliability check that determines whether each function's model is accurate enough to act on; a classification of each function into one of four evidence states, each with a defined action; and a decision log recording every choice, its rationale and its outcome. All code, data, results and weekly documentation are maintained in a public repository, so the full search is reproducible end to end.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

Key changes since the early rounds

The approach developed in three phases. In the opening rounds, a single exploration rule was applied uniformly across all eight functions. This was replaced quickly: the functions proved to have materially different behaviours, and each was assigned a strategy matched to its character.

The middle rounds expanded the toolkit and exposed its limits. In week 4, a model predicted a value of 2.14 in an unsampled region of function 4; the actual return was minus 17.9. That failure prompted the most important addition of the project: a weekly reliability check in which each known data point is withheld in turn and the model is asked to predict it. Any model that predicts held-out points worse than a simple average is excluded from decision-making. The check revealed that three of the eight functions had models that were actively misleading.

The later rounds consolidated. Small-step refinement near confirmed results replaced speculative search, and two deliberate reproduction tests (weeks 3 and 10) established that identical inputs return identical outputs to at least twelve decimal places. Every confirmed result is therefore repeatable on demand.

What influenced the changes

Portal results were the primary driver, followed by model diagnostics. Two external inputs also shaped the approach: the reliability check was adapted from a peer's published diagnostic, and the partition method trialled in week 7 implemented a published competition entry (NeurIPS 2020).

Principles now governing query decisions

Four rules apply. Models are acted on only after passing the reliability check. Incremental steps near confirmed results are preferred to long-range predictions, which have consistently underperformed. A verified baseline is secured before any budget is allocated to exploration. Every decision is documented with its rationale and outcome.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

The most significant trends

The strongest single pattern across twelve weeks of data is the relationship between query outcome and distance from the best known point. Queries placed close to a confirmed result return values close to it; queries placed at distance return materially worse values. All five improvements recorded in the last month came from steps within 0.02 of an existing best. Every long-range, prediction-led query since week 4 has underperformed its forecast.

The variables and behaviours that drive results

Function behaviour, rather than any single input variable, is the dominant factor, and it falls into four types. Two functions are highly sensitive: on function 4, a step of one hundredth away from the optimum reduced the output from 0.540 to 0.309. Two are tolerant: functions 6 and 7 have stable high-performing regions that have rewarded successive small steps with five improvements in four weeks. One is sparse: function 1 returns near-zero almost everywhere, with a single active region now located to within a few hundredths. One is resolved: function 5 has a verified maximum of 8,662 at an extreme corner, with all other tested regions returning considerably smaller values.

Within function 8, which has eight inputs, three inputs account for nearly all observed output variation. This is consistent across three independent measures of input importance; two inputs show no measurable effect at all.

How this shaped the understanding of the search

Search performance depends less on model sophistication than on correctly classifying each function's behaviour and matching the strategy to it. A step size that is safe on a tolerant function is costly on a sensitive one; exploration that is wasted on a resolved function is essential on one with an unresolved signal. The classification, not the model, became the core of the method.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing exploration and exploitation

The balance is set by evidence state rather than by a fixed schedule. Each week, every function is classified into one of four states, each with a defined action. Verified: the best result is confirmed and repeatable, so the confirmed coordinates are held back for the final submission and the weekly query is freed up for further exploration at no risk. Improving: small refinement steps near the current best, while the region continues to reward them. Localising: directional probes to pin down a signal whose position is only partly known. No reliable signal: the model has failed its reliability check, so queries are limited to simple, low-cost probes rather than model-led choices. Early rounds were weighted heavily toward exploration because little was known; the closing rounds are weighted toward exploitation because the evidence base now supports it.

Two decisions that illustrate the approach

The most successful decision combined model output with independent evidence. The brief described function 5 as having a single peak; the model repeatedly indicated the search-space boundary. Submitting the extreme corner in week 2 raised the best result from 1,089 to 8,662, and a repeat submission in week 3 returned the identical value, confirming the result.

The least successful decision relied on a model alone. In week 4, a prediction of 2.14 in an unsampled region of function 4 was acted on directly; the actual value was minus 17.9. The root cause was extrapolation without supporting data. The remediation was structural: the weekly reliability check now prevents any model that fails validation from directing a query, and predictions in unsampled regions are no longer acted on at any confidence level.

Handling uncertainty and unexpected results

Unexpected results are treated as diagnostic information. A sharp loss from a very small step on function 4 in week 9 identified the narrowness of that optimum and changed the final submission plan for that function. A change of sign between two nearby probes on function 1 in week 11 localised the target that ten weeks of broader searching had not found. In both cases the result contradicted expectations, and in both cases it materially improved the plan.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Planned actions for the final rounds

One search round remains before the final submission. The plan allocates it where evidence shows continuing improvement: function 1 (a directional step to complete the localisation of the active region), functions 6 and 7 (further small refinement steps in regions that continue to improve), and function 8 (a further application of the method that has delivered five consecutive improvements). The final round then submits the verified best coordinates for all eight functions. Because reproducibility has been established, the final values are known in advance with certainty.

Connection to the broader machine-learning landscape

The methodology used here - iterative modelling, uncertainty-aware selection, and strict evaluation budgets - is the same foundation that underpins commercial hyperparameter-optimisation services, experimental design in pharmaceutical and materials research, and large-scale testing platforms. The transferable element is the operating discipline: decisions gated on validated evidence, verification before reliance, and documentation sufficient for independent reproduction. These practices apply to any data-driven programme, regardless of the modelling technique in use.

Communicating results to a non-technical audience

The results would be presented as a one-page summary table: for each of the eight functions, the value at the start, the best value found, and the improvement achieved. Two or three headline examples would carry the story - for instance, function 5's output was improved eight-fold, from 1,089 to a verified 8,662. The method would be described in a single sentence: each week, the results gathered so far determined where the next test was placed. The closing message would address what a stakeholder most needs to know: every reported result has been re-tested and confirmed as repeatable, so the figures are reliable and can be acted on directly.